Chassis Physics and Space Planning

The Chassis Is the Robot's Foundation

The chassis is the base of the robot. It supports the robot's weight, holds the wheels and motors, protects the electronics and battery, and creates space for future mechanisms.

A good chassis is not just strong. It also needs to be useful. The shape of the chassis affects how the robot drives, turns, carries weight, handles contact, and makes room for attachments like a shooter, hopper, trigger, or intake.

The Starting Size Constraint

FTC robots usually need to fit inside an **18 inch x 18 inch x 18 inch** starting size. This means the robot must begin the match inside that size limit.

That does not mean every robot should use the full space. A larger robot may have more room for mechanisms, but a smaller or narrower robot may move through tight spaces, park, or share the field more easily.

For the DECODE-style challenge in this course, teams should think about:

- Can the robot fit at the start?
- Can it park or share space with other robots?
- Can game pieces move around, into, or through the chassis?
- Is there room for the battery, electronics, shooter, hopper, and trigger?
- Can the team reach important parts for repair?

Supporting Loads

The chassis carries everything else on the robot. If the frame bends, twists, or comes loose, every other system becomes less reliable.

The chassis needs to support:

- motors,
- wheels,
- battery,
- control hub and electronics,
- wires,
- shooter or scoring mechanism,
- hopper or game-piece storage,
- and any extra brackets or attachments.

Student-friendly idea:

A chassis is like a table. If the table wobbles, anything built on top of it becomes harder to use.

Staying Connected

A robot frame is made from parts connected by screws, nuts, brackets, plates, and connectors. The parts need to stay connected while the robot drives, turns, stops, bumps into objects, and carries weight.

Common chassis problems include:

- screws loosening,
- brackets bending,
- frame corners twisting,
- wheels rubbing,
- wires getting pulled,
- attachments shaking loose,
- or the battery sliding around.

Good teams check connections after testing. A robot that works once but falls apart after five minutes is not reliable yet.

Withstanding Contact

Robots may bump into game objects, field walls, or other robots. The chassis should be strong enough to handle normal contact without losing shape.

Questions to ask:

- Are important parts protected?
- Are wires away from wheels and moving parts?
- Are battery and electronics secure?
- Are corners strong enough?
- If the robot pushes a ball or artifact, does the frame twist?
- If another robot bumps us, will our mechanism stay aligned?

This does not mean the robot must be heavy or overbuilt. It means the frame should be strong enough for the job.

Motor Placement and Turning

Where the drive motors and drive wheels are placed changes how the robot feels to drive.

If the drive wheels are closer to the middle of the robot, the robot may turn more easily and may be able to rotate in place more smoothly. This can make the robot easier for a human driver to control.

If the drive wheels are farther apart or closer to the ends, the robot may feel more stable, but it may take more space to turn. It may also be harder to line up quickly.

Motor placement affects:

- turning radius,
- ability to rotate in place,
- how easy the robot is to control,
- how much space remains inside the frame,
- where the battery and electronics can go,
- how easy it is to add attachments later,
- and how protected the motors are.

There is no perfect motor placement. Teams need to decide what matters most: control, space, strength, repair access, or future mechanisms.

Driving in Multiple Directions

Most course robots will use two powered drive wheels and two support wheels, often omni wheels. This kind of robot can drive forward and backward and turn. It usually cannot slide sideways like a competition robot with mecanum wheels.

That means the chassis shape and motor placement matter. If the robot cannot move sideways, it must turn and line up carefully. A driver needs enough control to approach the goal, reach the human operator, avoid other robots, and park.

Clearance Off the Floor

Clearance means how much space there is between the bottom of the robot and the floor.

Too little clearance can cause:

- the robot to scrape,
- screws or brackets to catch on field elements,
- wheels to lose contact,
- or game pieces to jam under the frame.

Too much clearance can raise the robot's weight and make it less stable.

Good teams check whether the robot can move over the field surface without dragging or catching.

Space for Attachments

The chassis should leave room for future systems. A team may need space for:

- shooter,
- hopper,
- trigger servo,
- ball or artifact path,
- wiring,
- battery removal,
- electronics access,
- and repairs.

An open chassis may leave space for game pieces to enter or move through. A closed chassis may be stronger and easier to build. This is a trade-off.

Could Challenge: Add More Physics

If your team is ready for a challenge, you can explain the chassis using more formal physics ideas:

- **center of mass:** where the robot's weight is balanced,
- **torque:** turning force from motors or from weight placed far from the center,
- **friction:** grip between wheels and floor,

- **normal force:** how weight is shared across wheels,
- **stability:** how hard it is to tip the robot,
- **turning radius:** the space needed to complete a turn,
- **structural rigidity:** how much the frame resists bending or twisting.

You do not need these terms to build a first chassis, but they can help explain why one design works better than another.

Big Idea

The chassis is a set of engineering decisions. Shape, size, motor placement, clearance, and open space all affect how the robot works. A good chassis supports the robot, survives normal contact, leaves room for future mechanisms, and is easy enough to repair and improve.

Think About It

What should your team prioritize in the chassis: strength, open space, turning control, repair access, or room for future mechanisms?

____

____